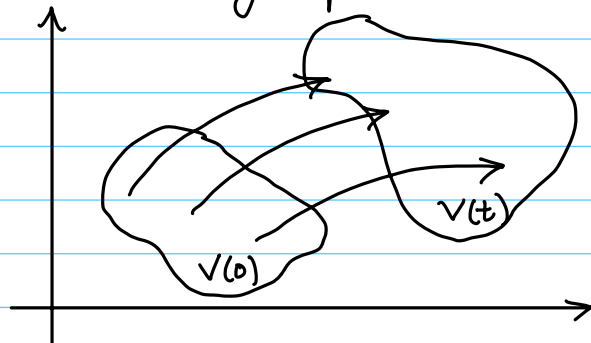


Math Thermo
 class #20
 04/07/2026

Theorem (Reynolds Transport Theorem) (20.1): Suppose that $V(t) \subseteq \Omega$ is a subset of Ω that moves according to velocity field $\vec{w}$



(20.1) Then,

$$\frac{d}{dt} \int_{V(t)} \tilde{q} d\vec{x} = \int_{V(t)} \left\{ \frac{\partial \tilde{q}}{\partial t} + \nabla \cdot (\tilde{q} \vec{w}) \right\} d\vec{x}$$

If $\vec{w}$ is the barycentric velocity, $\vec{v}$, then

$$\begin{aligned} \frac{d}{dt} \int_{V(t)} \rho g d\vec{x} &= \int_{V(t)} \left\{ \frac{\partial (\rho g)}{\partial t} + \nabla \cdot (\rho g \vec{v}) \right\} d\vec{x} \\ &= \int_{V(t)} \rho \frac{Dg}{Dt} d\vec{x}. \end{aligned}$$

Proof: See, for example,

Gonzalez and Stuart "A First Course in Continuum Mechanics," Cambridge (2008).

Example (20.2): let us apply the Reynold Transport Theorem to the mass density. Recall the continuity equation:

$$(20.2) \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

Assume $V(t)$ moves with velocity $\vec{v}$

RTT says that

$$\frac{d}{dt} M_{V(t)} = \frac{d}{dt} \int_{V(t)} \rho \, d\vec{x}$$

$$\stackrel{\text{RTT}}{=} \int_{V(t)} \left\{ \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) \right\} d\vec{x}$$

$$\stackrel{(19.2)}{=} 0.$$

Definition (20.3): Let $\rho_i : \Omega \times [0, T] \rightarrow [0, \infty)$ be the mass density of the i^{th} component in a mixture of r components. The total mass, as usual, is defined as

$$\rho := \sum_{i=1}^r \rho_i.$$

The quantity $\phi_i : \Omega \times [0, T] \rightarrow [0, 1]$, defined as

$$(20.3) \quad \phi_i := \frac{\rho_i}{\rho}, \quad (\text{unitless})$$

is called the mass fraction of component i .

Clearly,

$$\sum_{i=1}^r \phi_i = 1$$

and

$$\phi_i \geq 0$$

Recall that from (19.4) we have

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\rho_i \vec{v}) = -\nabla \cdot \vec{J}_i^D + \sigma_i$$

From our last definition we have

$$\rho_i = \rho \phi_i$$

Therefore,

$$(20.4) \quad \frac{\partial (\rho \phi_i)}{\partial t} + \nabla \cdot (\rho \phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D + \sigma_i$$

Using the material derivative, we find

$$(20.5) \quad \rho \frac{D\phi_i}{Dt} = -\nabla \cdot \vec{J}_i^D + \sigma_i.$$

Chemical Reactions

Chemical reactions can be modeled via the source terms. Let us suppose that there are K reactions. Set

$$(20.6) \quad \sigma_i = \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k, \quad 1 \leq i \leq r.$$

where $\tilde{\Gamma}_k$ is the reaction rate of the k^{th} reaction. Now, assume that

$$(20.7) \quad \sum_{i=1}^r \nu_{i,k} = 0, \quad 1 \leq k \leq K.$$

Then

$$\begin{aligned} \sum_{i=1}^r \sigma_i &= \sum_{i=1}^r \sum_{k=1}^K \nu_{i,k} \tilde{\Gamma}_k \\ &= \sum_{k=1}^K \tilde{\Gamma}_k \sum_{i=1}^r \nu_{i,k} \\ &= 0. \end{aligned}$$

The coefficients $\nu_{i,k}$ are related to the stoichiometric coefficients of the associated reactions. We'll talk about this later.

Conservation of Momentum

The momentum density is

$$\rho \vec{v} = [\rho v_1, \rho v_2, \rho v_3]^T$$

Then, momentum conservation is guaranteed by

$$(20.8) \quad \frac{\partial(\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \otimes \vec{v}) = -\nabla \cdot \mathcal{S} + \rho \vec{F},$$

where $\mathcal{S} \in \mathbb{R}^{3 \times 3}$ is the Cauchy stress tensor and $\vec{F}$ is the force density, per unit mass.

This law can be derived from the component form of the momentum conservation: $i=1, 2, 3$,

$$\text{or } \frac{\partial}{\partial t}(\rho v_i) + \nabla \cdot (\rho v_i \vec{v}) = -[\nabla \cdot \mathcal{S}]_i + \rho F_i,$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho v_i) + \sum_{j=1}^3 \frac{\partial}{\partial x_j}(\rho v_i v_j) \\ = - \sum_{j=1}^3 \frac{\partial}{\partial x_j} \mathcal{S}_{ij} + \rho F_i \end{aligned}$$

Since

$$[\vec{v} \otimes \vec{v}]_{ij} = v_i v_j,$$

$$\begin{aligned} \frac{\partial}{\partial t}(\rho v_i) + \sum_{j=1}^3 \frac{\partial}{\partial x_j}(\rho (\vec{v} \otimes \vec{v})_{ij}) \\ = - \sum_{j=1}^d \frac{\partial}{\partial x_j} \mathcal{S}_{ij} + \rho F_i \end{aligned}$$

This last equation is equivalent to (20.8).

$$[\vec{F}] = \frac{\text{Force}}{\text{mass}} = \frac{\text{Newtons}}{\text{kg}}$$

$$[\rho \vec{v}] = \frac{\text{mass} \cdot \text{velocity}}{\text{Volume}}$$

$$= \frac{\text{kg} \cdot \frac{\text{m}}{\text{sec}}}{\text{m}^3}$$

$$= \frac{\text{Newton} \cdot \text{sec}}{\text{m}^3}$$

$$\left[\frac{\partial(\rho \vec{v})}{\partial t} \right] = \frac{\text{Newton}}{\text{m}^3}$$

$$[\rho \vec{F}] = \frac{\text{Newtons}}{\text{m}^3}$$

Recall

$$1 \text{ Newton} = \frac{1 \text{ kg} \cdot \text{m}}{\text{sec}^2}$$

What are the units of the stress tensor?
Observe that

$$[\nabla \cdot \mathcal{S}] = \frac{\text{Newtons}}{\text{m}^3}$$

Thus,

$$[\mathcal{S}] = \frac{\text{Newtons}}{\text{m}^2}$$

$$= \text{Pascals} \quad (\text{units of pressure})$$

Recall

$$1 \text{ Pascal} = \frac{1 \text{ Newton}}{\text{m}^2}$$

The momentum conservation equation can be rewritten as

$$(20.9) \quad \rho \frac{D\vec{v}}{Dt} = - \underbrace{\nabla \cdot \mathcal{J}}_{\substack{\text{momentum density} \\ \text{diffusion}}} + \underbrace{\rho \vec{F}}_{\substack{\text{momentum density} \\ \text{source}}}$$

We can also express this equation as

$$(20.10) \quad \frac{\partial (\rho \vec{v})}{\partial t} + \nabla \cdot \mathcal{J} = \rho \vec{F}$$

where

$$\mathcal{J} = \rho \vec{v} \otimes \vec{v} + \mathcal{J} \in \mathbb{R}^{3 \times 3}$$

is the momentum flux tensor.

Using the Reynolds Transport Theorem and Divergence Theorem,

$$\begin{aligned} \frac{d}{dt} \int_{V(t)} \rho \vec{v} d\vec{x} &= \int_{V(t)} \rho \frac{D\vec{v}}{Dt} d\vec{x} \\ &\stackrel{(20.9)}{=} \int_{V(t)} \{ -\nabla \cdot \mathcal{J} + \rho \vec{F} \} d\vec{x} \end{aligned}$$

$$= - \int_{\partial V(t)} \mathcal{J} \hat{n} \, ds + \int_{V(t)} \rho \vec{F} \, d\vec{x},$$

where $\hat{n}$ is the outward-pointing unit normal on $\partial V(t)$.

The conservation of angular momentum implies that

$$\mathcal{J}^T = \mathcal{J},$$

ie, the Cauchy stress tensor is symmetric. To see a proof of this, see, again, Gonzalez and Stuart.

Conservation of Energy

let us start with the momentum conservation equation

$$(20.11) \quad \frac{\partial(\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \otimes \vec{v} + \mathcal{J}) = \rho \vec{F}.$$

let us assume that

$$\rho \vec{F} = \sum_{i=1}^r \rho_i \vec{F}_i, \quad (\vec{F} = \sum_{i=1}^r \phi_i \vec{F}_i)$$

where

$$\vec{F}_i = -\nabla \psi_i, \quad \frac{\partial \psi_i}{\partial t} = 0.$$

Define the mass-averaged potential energy, ψ , via

$$\rho \psi := \sum_{i=1}^r \rho_i \psi_i. \quad (\psi = \sum_{i=1}^r \phi_i \psi_i)$$

Then, dotting with $\vec{v}$,

$$\frac{\partial(\rho \vec{v})}{\partial t} \cdot \vec{v} + \vec{v} \cdot \nabla \cdot (\rho \vec{v} \otimes \vec{v} + \mathcal{J}) = \rho \vec{v} \cdot \vec{F}.$$

Now, observe that (exercise)

$$\begin{aligned} \rho \frac{D}{Dt} \left(\frac{1}{2} |\vec{v}|^2 \right) &= \rho \vec{v} \cdot \frac{D\vec{v}}{Dt} \\ &= \frac{\partial}{\partial t} (\rho \vec{v}) \cdot \vec{v} + \vec{v} \cdot \nabla \cdot (\rho \vec{v} \otimes \vec{v}). \end{aligned}$$

Therefore,

$$\rho \frac{D}{Dt} \left(\frac{1}{2} |\vec{v}|^2 \right) = -\vec{v} \cdot (\nabla \cdot \mathcal{J}) + \rho \vec{F} \cdot \vec{v}$$

or

$$(20.12) \quad \frac{\partial}{\partial t} \left(\frac{1}{2} \rho |\vec{v}|^2 \right) + \nabla \cdot \left(\frac{1}{2} \rho |\vec{v}|^2 \vec{v} \right) = -\vec{v} \cdot (\nabla \cdot \mathcal{J}) + \rho \vec{F} \cdot \vec{v}$$

Next we observe that

$$(20.13) \quad \vec{v} \cdot (\nabla \cdot \mathcal{J}) = \nabla \cdot (\mathcal{J}^T \vec{v}) - \mathcal{J} : \nabla \vec{v}.$$

Thus, combining the last two equations, we have

$$(20.14) \quad \boxed{\begin{aligned} \frac{\partial}{\partial t} \left(\frac{1}{2} \rho |\vec{v}|^2 \right) + \nabla \cdot \left(\frac{1}{2} \rho |\vec{v}|^2 \vec{v} + \mathcal{J}^T \vec{v} \right) \\ = \mathcal{J} : \nabla \vec{v} + \rho \vec{F} \cdot \vec{v} \end{aligned}}$$

let us prove (20.13), before we move on.

Proposition (20.4): The following identity holds:

$$(20.13) \quad \vec{v} \cdot (\nabla \cdot \mathcal{J}) = \nabla \cdot (\mathcal{J}^T \vec{v}) - \mathcal{J} : \nabla \vec{v}.$$

Proof: $\mathcal{J}^T \vec{v}$ is a vector with components

$$[\mathcal{J}^T \vec{v}]_i = \sum_{j=1}^3 \mathcal{J}_{ji} v_j = \mathcal{J}_{ji} v_j,$$

using Einstein's summation convention. Thus

$$\begin{aligned} \nabla \cdot (\mathcal{J}^T \vec{v}) &= \sum_{i=1}^3 \frac{\partial}{\partial x_i} \left(\sum_{j=1}^3 \mathcal{J}_{ji} v_j \right) \\ &= \sum_{i=1}^3 \sum_{j=1}^3 \frac{\partial}{\partial x_i} (\mathcal{J}_{ji} v_j) \\ &= \frac{\partial}{\partial x_i} (\mathcal{J}_{ji} v_j) \\ &= \left(\frac{\partial}{\partial x_i} \mathcal{J}_{ji} \right) v_j + \mathcal{J}_{ji} \frac{\partial v_j}{\partial x_i} \\ &= (\nabla \cdot \mathcal{J}) \cdot \vec{v} + \mathcal{J} : \nabla \vec{v}. \quad /// \end{aligned}$$

Here we use the notation

$$A : B = \sum_{i=1}^d \sum_{j=1}^d A_{ij} B_{ij} = A_{i,j} B_{i,j}.$$

Next, let us derive an equation for the balance of potential energy.

The barycentric potential energy is defined via

$$\rho \psi = \sum_{i=1}^r \rho_i \psi_i.$$

Thus,

$$\psi = \sum_{i=1}^r \phi_i \psi_i \quad (\text{specific pot. energy density})$$

By properties of the material derivative,

$$\begin{aligned} \frac{\partial}{\partial t} (\rho \psi) + \nabla \cdot (\rho \psi \vec{v}) &= \rho \frac{D\psi}{Dt} \\ &= \rho \sum_{i=1}^r \left[\phi_i \frac{D\psi_i}{Dt} + \psi_i \frac{D\phi_i}{Dt} \right]. \end{aligned}$$

Using the mass conservation equation (20.5)

$$\begin{aligned} \rho \frac{D\phi_i}{Dt} &= -\nabla \cdot \vec{J}_i^D + \sigma_i \\ &= -\nabla \cdot \vec{J}_i^D + \sum_{k=1}^K \nu_{i,k} \tilde{\sigma}_k, \end{aligned}$$

we have

$$\begin{aligned} \frac{\partial}{\partial t} (\rho \psi) + \nabla \cdot (\rho \psi \vec{v}) &= \rho \sum_{i=1}^r \phi_i \cancel{\frac{\partial \psi_i}{\partial t}} + \rho \sum_{i=1}^r \phi_i \vec{v} \cdot \nabla \psi_i \\ &\quad + \sum_{i=1}^r \psi_i \rho \frac{D\phi_i}{Dt} \end{aligned}$$

$$= \rho \sum_{i=1}^r \phi_i \vec{v} \cdot \nabla \psi_i - \sum_{i=1}^r \psi_i \nabla \cdot \vec{J}_i^D$$

$$+ \sum_{i=1}^r \psi_i \sum_{k=1}^K v_{i,k} \hat{\sigma}_k$$

We assume that, for each $k=1, \dots, K$,

$$\sum_{i=1}^r \psi_i v_{i,k} = 0,$$

Since, generally, the potential energy is unchanged by a chemical reaction.

Thus,

$$\frac{\partial}{\partial t} (\rho \psi) + \nabla \cdot (\rho \psi \vec{v})$$

$$= \rho \sum_{i=1}^r \phi_i \vec{v} \cdot \nabla \psi_i - \sum_{i=1}^r \psi_i \nabla \cdot \vec{J}_i^D$$

$$= -\rho \sum_{i=1}^r \phi_i \vec{v} \cdot \vec{F}_i - \sum_{i=1}^r \psi_i \nabla \cdot \vec{J}_i^D$$

$$= -\rho \vec{F} \cdot \vec{v} - \sum_{i=1}^r \psi_i \nabla \cdot \vec{J}_i^D.$$

Note that

$$\nabla \cdot (\psi_i \vec{J}_i^D) = \nabla \psi_i \cdot \vec{J}_i^D + \psi_i \nabla \cdot \vec{J}_i^D.$$

Thus,

$$-\psi_i \nabla \cdot \vec{J}_i^D = \nabla \psi_i \cdot \vec{J}_i^D - \nabla \cdot (\psi_i \vec{J}_i^D)$$

and we obtain

$$\begin{aligned} \frac{\partial}{\partial t} (\rho \psi) + \nabla \cdot (\rho \psi \vec{v}) \\ = -\rho \vec{F} \cdot \vec{v} + \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D - \sum_{i=1}^r \nabla \cdot (\psi_i \vec{J}_i^D) \end{aligned}$$

Finally, we have

(20.15)

$$\begin{aligned} \frac{\partial}{\partial t} (\rho \psi) + \nabla \cdot (\rho \psi \vec{v} + \sum_{i=1}^r \psi_i \vec{J}_i^D) \\ = -\rho \vec{F} \cdot \vec{v} + \sum_{i=1}^r \nabla \psi_i \cdot \vec{J}_i^D. \end{aligned}$$

This is the conservation equation for potential energy.